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SATHYABAMA
INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CONTINUOUS ASSESSMENT EXAM-2

Programme:B.E-ECE
Course Code: SECB1301
Course Name: Electronic Circuits
Batch : 2024-2028

Max. Marks: 50
Time: 2 Hours
Sem: III
Date:08/10/2025

Part-A Answer ALL the questions (7 × 2=14)

Q.No	Questions	CO
1.	Compare positive feedback and negative feedback.	3
2.	Determine the gain of an amplifier if the feedback gain is 75 and feedback factor is 0.01.	3
3.	Specify the advantages of negative feedback.	3
4.	Sketch the general structure of feedback amplifier.	3
5.	Calculate the frequency of oscillation for a Hartley oscillator if L1=2.3mH,L2=23mH and C=0.01μf.	4
6.	State the barkhausen criterion for oscillation.	4
7.	Specify an examples of high frequency oscillator.	4

Part-B Answer ALL the questions (3 × 12 = 36)

8.	Determine the gain, input impedance and output impedance of voltage shunt feedback connection.	3
	(OR)	
9.	Enumerate the effects of negative feedback on distortion, noise and stability.	3
10.	Determine the gain, input impedance and output impedance of current series feedback connection.	3
	(OR)	
11.	Interpret the frequency of oscillation and condition of oscillation for a Hartley oscillator.	4
12.	Examine the general equation of LC oscillator with equivalent circuit.	4
	(OR)	
13.	Interpret the frequency of oscillation and condition of oscillation for a colpitts oscillator.	4